

Exercise series 7:

FDs & BCNF



ADReM
Advanced Database Research & Modelling
University of Antwerp

 Universiteit
Antwerpen

Find All the Keys

R(A, B, C, D, E)

- $A, B \rightarrow E$
- $E \rightarrow A, B$
- $D \rightarrow C$



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- $A, B \rightarrow E$
- $E \rightarrow A, B$
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Initialize:

- Attribute not on the right = $\{D\}$
- Closure of those attributes = $\{D\}^+ = \{C, D\}$
- Subsets of the remaining attributes = $\mathcal{P}(A, B, E) = \{\}, \{A\}, \{B\}, \{E\}, \{A, B\}, \{A, E\}, \{B, E\}, \{A, B, E\}$

Found Keys =



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Set to consider:

$\{\}, \{A\}, \{B\}, \{E\}, \{A, B\}, \{A, E\}, \{B, E\}, \{A, B, E\}$

- $\{D\} \cup \{\} = \{D\}$
- $\{D\}^+ = \{C, D\}$
- **NOT a Key**

Found Keys =



Find All the Keys

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Set to consider: $\{\}, \{A\}, \{B\}, \{E\}, \{A, B\}, \{A, E\}, \{B, E\}, \{A, B, E\}$

- $\{D\} \cup \{A\} = \{A, D\}$
- $\{A, D\}^+ = \{A, C, D\}$
- **NOT a Key**

Found Keys =



Find All the Keys

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Set to consider: $\{\}, \{A\}, \{B\}, \{E\}, \{A, B\}, \{A, E\}, \{B, E\}, \{A, B, E\}$

- $\{D\} \cup \{B\} = \{B, D\}$
- $\{B, D\}^+ = \{B, C, D\}$
- **NOT a Key**

Found Keys =



Find All the Keys

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Set to consider: $\{\}, \{A\}, \{B\}, \{E\}, \{A, B\}, \{A, E\}, \{B, E\}, \{A, B, E\}$

- $\{D\} \cup \{E\} = \{D, E\}$
- $\{D, E\}^+ = \{A, B, C, D, E\}$
- **KEY FOUND**

Found Keys = $\{D, E\}$



Find All the Keys

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- $A, B \rightarrow E$
- $E \rightarrow A, B$
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- Subsets of the remaining attributes = $\mathcal{P}\{A, B, E\} = \{\}, \{A\}, \{B\}, \{E\}, \{A, B\}, \{A, E\}, \{B, E\}, \{A, B, E\}$

Set to consider: $\{\}, \{A\}, \{B\}, \{E\}, \{A, B\}$

- $\{D\} \cup \{A, B\} = \{A, B, D\}$
- $\{A, B, D\}^+ = \{A, B, C, D, E\}$
- **KEY FOUND**

Found Keys = $\{D, E\}, \{A, B, D\}$



Find All the Keys

R(A, B, C, D, E)

- $A, B \rightarrow E$
- $E \rightarrow A, B$
- $D \rightarrow C$

Initialize:

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- Closure of those attributes = $\{D\}^+ = \{C, D\}$
- Subsets of the remaining attributes = $\mathcal{P}\{A, B, E\} = \{\}, \{A\}, \{B\}, \{E\}, \{A, B\}, \{A, E\}, \{B, E\}, \{A, B, E\}$

Set to consider: $\{\}, \{A\}, \{B\}, \{E\}, \{A, B\}$

FINAL KEYS

Found Keys = $\{D, E\}, \{A, B, D\}$



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BCNF decomposition

- $A, B \rightarrow E$
- $D \rightarrow A, B$
- $E \rightarrow C$

$R(A, B, C, D, E)$

BCNF decomposition

- $A, B \rightarrow E$
- $D \rightarrow A, B$
- $E \rightarrow C$



- $A, B \rightarrow E$
- $\{A, B\}^+ = \{A, B, C, E\}$

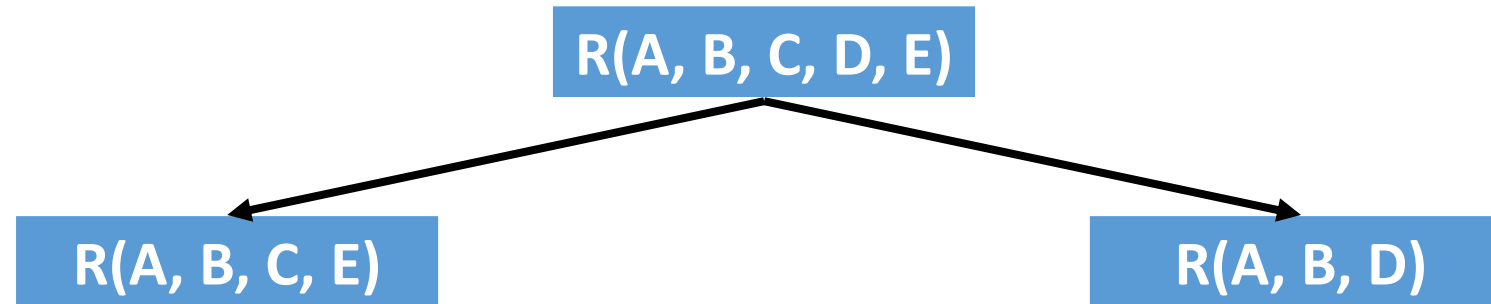
$R(A, B, C, D, E)$

BCNF decomposition

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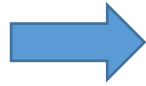


- $A, B \rightarrow E$
- $\{A, B\}^+ = \{A, B, C, E\}$

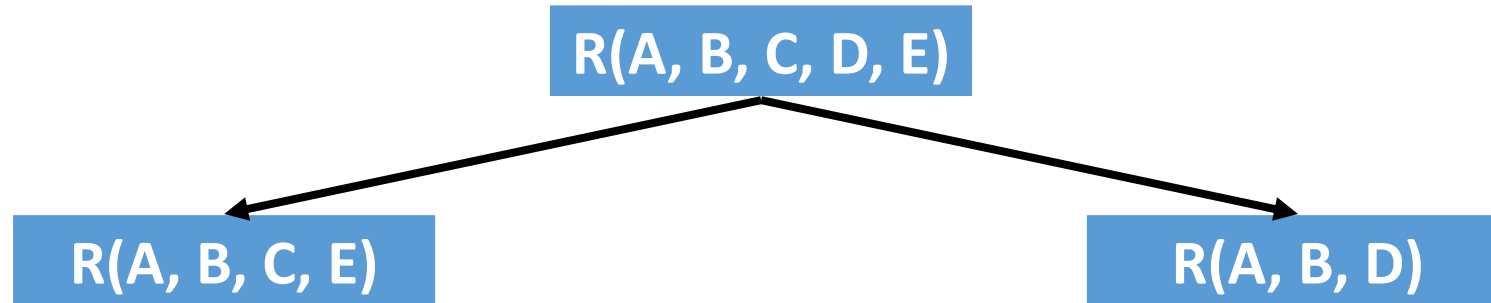


BCNF decomposition

- $A, B \rightarrow E$
- $D \rightarrow A, B$
- $E \rightarrow C$



- $E \rightarrow C$
- $\{E\}^+ = \{C, E\}$

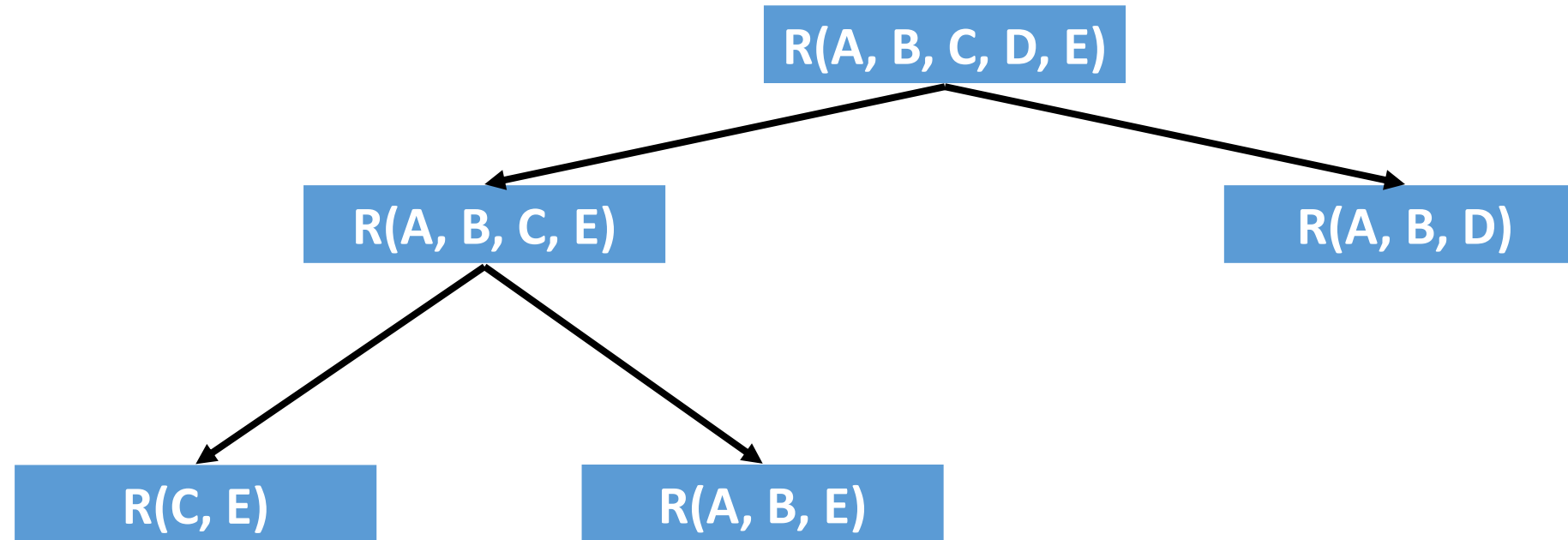


BCNF decomposition

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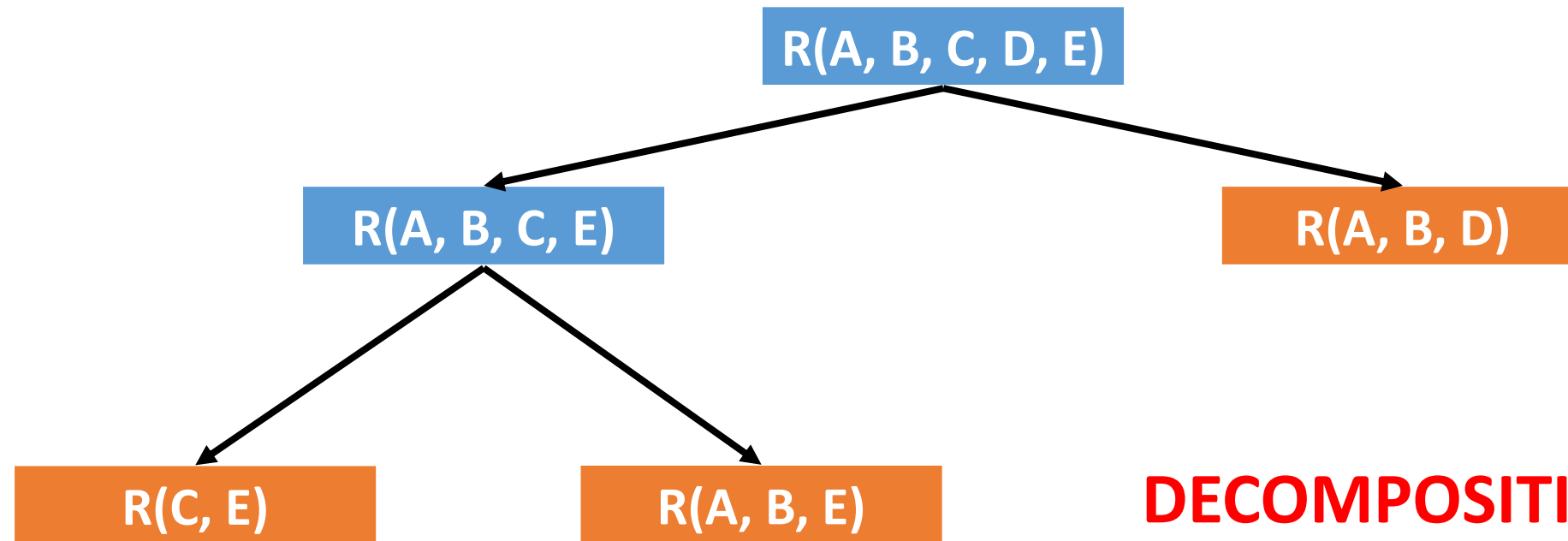


- $E \rightarrow C$
- $\{E\}^+ = \{C, E\}$



BCNF decomposition

- $A, B \rightarrow E$
- $D \rightarrow A, B$
- $E \rightarrow C$



DECOMPOSITION FOUND

Is the decomposition Lossless?

R(A, B, C, D)

- $A \rightarrow C$
- $D \rightarrow A$

Decomposition

- $R_1(B, D)$
- $R_2(A, C)$
- $R_3(A, D)$



Is the decomposition Lossless?

R(A, B, C, D)

- $A \rightarrow C$
- $D \rightarrow A$

- Create table

Decomposition

- $R_1(B, D)$
- $R_2(A, C)$
- $R_3(A, D)$

	A	B	C	D
R_1				
R_2				
R_3				



Is the decomposition Lossless?

$R(A, B, C, D)$
$A \rightarrow C$ $D \rightarrow A$

- Create table
- Populate table

Decomposition
• $R_1(B, D)$ • $R_2(A, C)$ • $R_3(A, D)$

	A	B	C	D
R_1	a_1	b	c_1	d
R_2				
R_3				



Is the decomposition Lossless?

R(A, B, C, D)

- $A \rightarrow C$
- $D \rightarrow A$

- Create table
- Populate table

Decomposition

- $R_1(B, D)$
- $R_2(A, C)$
- $R_3(A, D)$

	A	B	C	D
R_1	a_1	b	c_1	d
R_2	a	b_2	c	d_2
R_3				



Is the decomposition Lossless?

R(A, B, C, D)

- $A \rightarrow C$
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- Create table
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Decomposition

- $R_1(B, D)$
- $R_2(A, C)$
- $R_3(A, D)$

	A	B	C	D
R_1	a_1	b	c_1	d
R_2	a	b_2	c	d_2
R_3	a	b_3	c_3	d



Is the decomposition Lossless?

R(A, B, C, D)

- $A \rightarrow C$
- $D \rightarrow A$

- Create table
- Populate table
- Update table

Decomposition

- $R_1(B, D)$
- $R_2(A, C)$
- $R_3(A, D)$

	A	B	C	D
R_1	a_1	b	c_1	d
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	A	B	C	D
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Decomposition

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- $R_3(A, D)$

	A	B	C	D
R_1	a	b	c_1	d
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Is the decomposition Lossless?

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Decomposition

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- $R_2(A, C)$
- $R_3(A, D)$

	A	B	C	D
R_1	a	b	c_1	d
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R_2	a	b_2	c	d_2
R_3	a	b_3	c	d

Correct Row Found!

